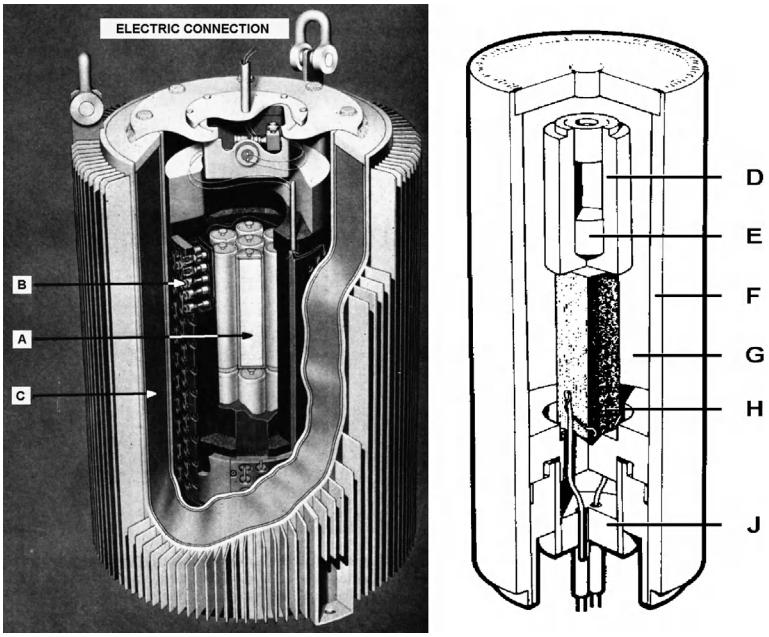


Figure 7.29 A - $^{90}SrTiO_3$; B - thermocouples; C - radiation shield (depleted U); D - capsule; E - ^{238}Pu; F - secondary containment; G - thermal insulation; H - thermopile; J - insulator.

450 W. One will reach the star Sirius in about 300 000 years! In 2012, the most recent RTG use as a power source in space vehicles is on the New Horizons mission to Pluto and Charon.

7.10. EXERCISES

7.1. In §7.3 two equations are given for calculating the range of α-particles in air and in other material from the particle energy as well as a curve for the range in aluminum. How different are the values from the equations and from the curve for a 5 MeV α?

7.2. What is the minimum energy that an α-particle must have to be detected by a GM tube having a mica window of 1.5 mg cm^{-2} (the density of mica is approximately equal to aluminum)?

7.3. For an irradiation experiment it is necessary to extract a beam of deuterons from an accelerator. The projectile energy is 22 MeV D^+. For this purpose the beam is deflected and permitted to pass through a thin titanium foil (density 4.5 g cm^{-3}). Assuming that $\hat{R}_1 \rho_1 M_1^{-½} = \hat{R}_2 \rho_2 M_2^{-½}$ (Bragg-Kleman rule), what is the maximum thickness of the foil? Give the answer in millimeters.

7.4. Make a rough estimate of the range in air for a 1 MeV α, 1 MeV H^+ and 1 MeV e^- using the plot in Fig. 7.7. The energy to form an ion pair in air is 14.6 eV but assume that twice as much energy is lost through excitation.

7.5. What is the range of a 6.3 MeV α-particle in (a) aluminum, (b) nickel, (c) platinum?

7.6. What is the γ-ray flux from a 3.7 GBq ^{60}Co source at a distance of 3 m? Assume $\psi_{\text{sample}} = 1$.

7.7. What is the maximum range in millimeters of β-particles from T, ^{14}C, ^{32}P and ^{90}Sr in a photographic emulsion if its absorption efficiency is assumed to be the same as aluminum? The density of the emulsion is assumed to be 1.5 g cm^{-3}.

7.8. The $E_{\max}$ of ^{32}P β-particles is 1.71 MeV. To what electron velocity does this correspond?

7.9. In a laboratory an irradiation area must be designed for γ-radiography using a 3.7×10^{11} Bq ^{60}Co source. For this purpose a cubic building is erected with an interior side length of 2 m. The desired flux reduction is 10^6. How thick must the wall be and how much will the shielding material cost (i.e. not including labor costs) if it is made of (a) concrete? (b) lead? Assume lead blocks cost \$1.50 per kg and concrete \$40 per m^3.

7.10. An experiment is done with ^{60m}Co which emits 0.05860 MeV γ. The detector used is a NaI crystal. What photo peaks will be observed if the electron binding energies in sodium are K 1072 and L 63 eV, and in iodine K 33 170 and L 4800 eV?

7.11. A human body may be considered as consisting of water. Radiation from ^{137}Cs is absorbed by a 15 cm thick body. How much is the γ-ray flux reduced by the body, and how much of the beam energy (β plus γ) is absorbed?

7.12. For iron the mass attenuation coefficients are: at 0.5 MeV γ, 0.083; at 1.0 MeV, 0.059; at 1.5 MeV, 0.047 cm^2 g^{-1}. Calculate the corresponding one-tenth values.

7.13. An absorption curve of a sample emitting β- and γ-rays was taken with aluminum absorber using a gas-flow proportional counter. The data obtained were:

Absorber thickness (g cm^{-2})	Activity (counts min^{-1})	Absorber thickness (g cm^{-2})	Activity (counts min^{-1})
0	5800	0.700	101
0.070	3500	0.800	100
0.130	2200	1.00	98
0.200	1300	2.00	92
0.300	600	4.00	80
0.400	280	7.00	65
0.500	120	10.00	53
0.600	103	14.00	40

(a) Estimate the maximum energy of the β-spectrum. (b) Find the energy of the γ-ray.

7.14. A 40-story high modern business building is supported by 0.9 m thick pillars of reinforced concrete. The insurance company must check that the number of iron bars are as many as required, and therefore they want to investigate the pillars by γ-radiography. What exposure times are required for (a) a small 200 GBq ^{60}Co source, (b) for a large 150 TBq source? Use the same film data as in 7.9.2.

7.15. A swimming-pool reactor produces a flux of 3×10^{16} thermal neutrons $m^{-2}\,s^{-1}$ at 1 m from the reactor center. Assuming a parallel beam of neutrons diffusing up to the surface of the pool where the neutron flux is measured to be 10^8 n $m^{-2}\,s^{-1}$, calculate the thickness (x m) of the water layer required. For thermal neutrons the flux is reduced exponentially with the exponent $x\,L^{-1}$, where L is the diffusion length (2.75 cm in H_2O).

7.16. In a sample of 10.4 TBq of old fission products, the average γ-ray energy is 0.5 MeV and on the average 0.4 γ's are emitted per β-decay. (a) What is the lead shielding required to reduce the γ flux to 10^2 γ $cm^{-2}\,s^{-1}$ at 1.5 m from the source assuming only exponential attenuation? (b) What is the relaxation length? (c) What is the build-up factor?

7.11. LITERATURE

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